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Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The client is DriverPass, a driving school company (specifically Liam, the owner/manager).
- The purpose of the project is to transition DriverPass from a manual/paper-based scheduling process to a computerized, digital system.
- They want a system that can manage student information, instructor schedules, and vehicle availability in one centralized location to prevent scheduling conflicts and improve efficiency.
- The system aims to improve student pass rates by providing a digital platform for online practice exams, addressing the high failure rate identified in DriverPass research.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass currently suffers from inefficiencies due to manual tracking, leading to potential double-booking or lost records.
- The system needs to allow administrators to input and retrieve data regarding students, instructors, and driving sessions.
- DriverPass needs to solve the issue of students being underprepared for written exams by offering accessible, remote practice testing.

Components needed:

- A database (to store student, instructor, and schedule data).

- A user interface (front-end) for easy interaction.
- Business logic (middle-ware) to handle the rules of scheduling (e.g., checking if a car is free).
- An Exam Module/Engine (to host, deliver, and automatically grade practice tests).

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Securely store and retrieve student and instructor records.
- Allow for the creation, modification, and deletion of driving appointments.
- Validate data entry to ensure no duplicate appointments are created.
- Provide a secure login for administrators to protect sensitive student data.
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Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system should be a web-based application to allow access from office computers without specialized software installation.
- The system should load student records within 2 seconds to ensure efficient workflow for the staff.
- The database should update in real-time immediately after a schedule change is submitted to prevent overlapping bookings.
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Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should be platform-independent regarding the client side (accessible via standard web browsers like Chrome or Edge on Windows or macOS).
- The back end will require a relational database management system (such as MySQL or PostgreSQL) to manage the data relationships.
- The application requires a server environment capable of hosting Java or similar web-application languages.
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Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system will assign a unique ID to every student and instructor to distinguish between users with similar names.
- Passwords shall be case-sensitive; usernames shall be case-insensitive.
- The system must inform the admin immediately (via error message) if they attempt to book an appointment that conflicts with an existing one.
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Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system must allow the IT admin to add, edit, or archive users (students/instructors) through the interface without needing to alter the source code or database manually.
- The system should be built using standard web technologies (HTML5/CSS) to remain compatible with future browser updates.
- The IT admin requires "Super User" access to manage lower-level system configurations and password resets.
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Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users are required to provide a valid username and password combination to log in.
- Data exchange should be secured using HTTPS/SSL encryption to protect student privacy during transmission.
- The account should be temporarily locked (e.g., for 15 minutes) after 3 consecutive failed login attempts to prevent brute force attacks.
- If a user forgets their password, the system should allow an administrator to generate a temporary password or reset link.
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Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall validate user credentials (username and password) against the database upon login.
- The system shall allow the user to create a new student profile containing name, contact info, and permit status.
- The system shall allow the user to schedule a driving session by selecting a student, an instructor, and a time block.
- The system shall verify that both the instructor and the vehicle are available before confirming a scheduled appointment.
- The system shall allow the user to view a daily or weekly schedule of all driving sessions.
- The system shall allow the user to update or cancel an existing appointment.
- The system shall allow students to log in and access a library of online practice exams.
- The system shall automatically grade student practice exams and provide immediate feedback on incorrect answers.
- The system shall allow administrators to upload, edit, or remove practice exam questions.
- The system shall track and store student exam scores for progress monitoring by instructors.
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User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- **Primary User:** The administrative staff/receptionist (and potentially Liam, the owner).
- **Administrator/Staff:** Needs the calendar view for scheduling and the ability to manage student/instructor records.
- **Student:** Needs a personalized dashboard to access study materials and launch practice exams with clear "Next" and "Submit" navigation.
- **Needs:** The interface needs to be clean, high-contrast, and intuitive, requiring minimal clicks to perform common tasks like checking a schedule. It also needs all students to have a personalized dashboard where they can access study materials, launch practice exams, and view their historical test scores
- **Interaction:** Users will interact via a desktop or laptop web browser using a mouse and keyboard. Students will also need to be able to interact via a standard web browser on a desktop, laptop, or tablet. The interface for exams must be

straightforward, featuring clear navigation buttons (e.g., "Next Question," "Submit Test").

- **Visualization:** A calendar view is essential for visualizing open slots versus booked slots. There should also be a progress bar or score chart that is necessary to help students visualize their improvement over time as they prepare for the official driving license exam.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- It is assumed that the office has a reliable internet connection to access the web-based system.
- It is assumed that the client (DriverPass) has existing data (student lists) that will need to be manually entered or imported into the new system.
- It is assumed that the staff has basic computer literacy skills to navigate a web browser.
- It is assumed that DriverPass will provide the initial bank of practice exam questions and correct answers for database entry.
- It is assumed that students have access to their own internet-connected devices (laptops, tablets) to utilize the remote practice testing features.
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Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- **Resource Limitation:** As this is a student project, there is only one developer, which limits the scale of features that can be implemented (e.g., no mobile app version at this time).
- **Budget Limitation:** The system relies on open-source or free-tier database solutions, which may have storage limits.
- **Design Limitation:** The system currently does not handle payment processing (credit cards), meaning payments must still be handled externally.
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Gantt Chart

Please include a screenshot of the GANTT chart that you created with by the characters in the interview.

. Be sure to check that it meets the plan described

DriverPass Schedule

TASK NAME	PHASE	START	END	Collect Requirements
Collect Requirements	ANALYSIS	Jan 22	Feb 4	Create Use Case Diagrams
Create Use Case Diagrams	DESIGN	Feb 11	Feb 18	Build Activity Diagrams
Build Activity Diagrams	DESIGN	Feb 15	Mar 9	Research UI Designs
Research UI Designs	DESIGN	Feb 27	Mar 7	Build Class Diagram
Build Class Diagram	DESIGN	Mar 1	Mar 9	Get Customer Approval
Get Customer Approval	MANAGEMENT	Mar 10	Mar 11	Build Interface
Build Interface	DEVELOPMENT	Mar 12	Mar 24	Link DB to Interface
Link DB to Interface	DEVELOPMENT	Mar 24	Apr 3	Build Business Logic
Build Business Logic	DEVELOPMENT	Apr 5	Apr 27	Test System
Test System	TESTING	Apr 27	May 7	Deliver System
Deliver System	DEPLOYMENT	May 8	May 9	Sign-off Meeting
Sign-off Meeting	MANAGEMENT	May 9	May 10	

Feb Mar Apr May